Evidence-Constrained Reconstruction Repair and Residual Humanoid Grounding

Extended Abstract

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We present our V2D Track 2 system across clean, corrupted and RGB-reconstructed human–object references. A common G1 pipeline combines retargeting, frozen SONIC body control, phase/proximity hand closing and optional episode-specific FlashSAC residual actors. Repairs are admitted using source evidence, object settling and unassisted physical evaluation. Selected 30-episode submissions achieve AUC 0.32917, 0.25073 and 0.19051 in Tiers 1–3, respectively. Both success rates reach 4/30 in Tier 1 and 1/30 in Tier 2, but remain zero in Tier 3. Our contribution is a reusable, auditable repair-to-control workflow and an empirical account of where better visual tracking fails to produce manipulation.

1. Introduction

Human demonstrations offer inexpensive supervision, but geometric uncertainty, morphology differences and physical contact complicate transfer to a robot. V2D [4] separates these effects using clean references (Tier 1), synthetically corrupted references (Tier 2) and participant reconstructions (Tier 3). All tiers are evaluated against clean object trajectories. We keep tier-specific references and learned actors separate; we do not transfer clean-trained episode actors into the noisy tiers.

We build on SONIC [2] and FlashCHORD/FlashSAC [1, 3]. Our work develops reconstruction repair, scene admission and controlled policy selection rather than a new foundation model. The selected files are Tier 1 Submission 7, Tier 2 Submission 2 and Tier 3 Submission 4.

2. Approach

Reconstruct and repair. Tier 3 uses a CARI4D-style initialization [5]: SAM2 masks, MoGe-2 depth, SAM 3D body/object estimates and FoundationPose object tracking. We repair incorrect object identity, incomplete masks, metric inconsistencies, occlusion recovery, rotation jumps and support geometry. Visible-frame anchors and bounded temporal fitting preserve accepted intervals and test joins. Image overlap alone cannot establish depth, contact or physical validity. Hidden intervals retain explicit uncertainty. The permitted first clean frame is used for alignment; later clean trajectories are reserved for evaluation, not reconstruction or actor observations. Episode-specific proposals and manual visual review remain part of this system.

Retarget and control. MHR-to-SOMA conversion and constrained inverse kinematics generate G1 reference motion. A portable scene includes reconstructed geometry, support, inertia and collision assets. Frozen SONIC controls the body. Each hand closes gradually during its reference contact phase, gated by actual palm proximity to a target attached to the current object pose. Targets obey rate and joint limits. Optional residual actors add bounded body and finger corrections; zero residual does not mean zero finger targets. Each sequence has its own policy assignment.

Train and admit. FlashSAC learns off-policy from bounded replay in parallel simulation. Rewards combine body, palm and object tracking, contact guidance and regularization; orientation and broader approach shaping are explicit experiment variants. Curricula mix starts at frame zero with starts partway through the reference. Early runs removed artificial object assistance around 6M transitions; recent pilots taper it to zero by 10M. All reported evaluations start at frame zero without assistance. Source checks, isolated object settling and repeated rollouts precede candidate admission. A checkpoint is not selected solely because training reward or training duration increases.

3. Experiments & Results

We used a 24 GB RTX 5090 laptop and bounded L4 cloud jobs. Local GPU clocks were capped at 1500 MHz; L4 power at 60 W. Recent jobs pause at 82 degrees C, resume at 74 degrees C and stop at 83 degrees C. Table 1 reports exact Kaggle scores for complete files, not cherry-picked repeats. AUC and success rates are higher-is-better; MPPE is lower-is-better in centimetres.

Tier	File	AUC	SP-SR	MP-SR	MPPE
1	Initial	.21646	.00000	.00000	56.78341
1	Selected S7	.32917	.13333	.13333	39.21800
2	Initial	.22281	.00000	.00000	59.14549
2	Selected S2	.25073	.03333	.03333	55.10178
3	Initial	.13370	.00000	.00000	70.87785
3	Selected S4	.19051	.00000	.00000	59.96399

Table 1: Each file covers all 30 episodes. Tier differences include both input quality and policy selection; this is not a controlled noise ablation.

Tier 1 gains success from residual learning and broader episode selection. Tier 2's selected Episode 0 actor passes both success criteria in five confirmation rollouts. Tier 3 S4 replaces only Episodes 20 and 26 with repaired frozen policies, retaining the other 28 designated trajectories on the same evaluator build. Its selected set includes six previously learned residual actors, but no episode passes the task-success criteria.

Negative results constrain interpretation. Episode 26's new 15M approach actor extends upright survival from 9.79 to 18.51 seconds and improves reaching, yet raises clean-target MPPE from approximately 7.96 to 9.45 cm. We retain the repaired frozen controller. New Episode 20/7 actors can also lose to retained policies despite better reference-relative diagnostics. Quieter object rest can improve AUC without grasping; we do not describe these gains as successful manipulation. Repeated rollouts measure execution variability, not variation across independently trained seeds.

4. Conclusion

Conservative repairs and selective residual learning improve object tracking, while sustained contact and task timing remain the main limitations. Our **economic viability** contribution is reuse of pretrained models, checkpointed bounded jobs and rejected-candidate early stopping; we have no audited end-to-end cost reduction claim. The **data flywheel** records failure evidence, accepted repairs and regression checks in reusable modules, although diagnosis still requires human review. **Complexity reduction** comes from explicit geometry/policy contracts and four separable modules: reconstruction, repair, retargeting and controller training. **Innovation** is testing visual and physical validity separately and retaining simpler controllers when learning regresses. These are demonstrated research mechanisms, not claims of autonomous production deployment.

Code: https://github.com/isyed/v2d-grounding; reusable modules: https://github.com/isyed/real2sim2real. Exact producing commits, submission IDs and reproduction status accompany this report. No real-robot evaluation or generalization beyond the challenge episodes is claimed.

References

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